

An aerial photograph of the University of Colorado Boulder campus. In the center is a large, historic red brick building with a prominent central tower and a flagpole on top. The building is surrounded by lush green trees with some autumn-colored foliage. In the background, a large, rugged mountain with rocky peaks rises under a blue sky with scattered clouds.

Beidou B1/B2 SDR

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University of Colorado **Boulder**

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Introduction

Introduction (1/2)

	GPS	Beidou
Constellation	MEO(24+ α)	GEO(5)+MEO(27)+IGSO(3)
Coordinate System	WGS84	CGCS2000
Time System	GPST	BDT (GPST = BDT + 14)
Frequency	1575.42 MHz	1561.098 MHz for B1 1207.14e6 MHz for B2
Modulation	BPSK	QPSK (NH code modulation for D1)
Code	1.023 MHz (1023 chips/code)	2.046 MHz (2046 chips/code)
Navigation Message	50 bps	D1 (50 bps) & D2 (500 bps)

Introduction (2/2)

■ Status of Beidou SVs

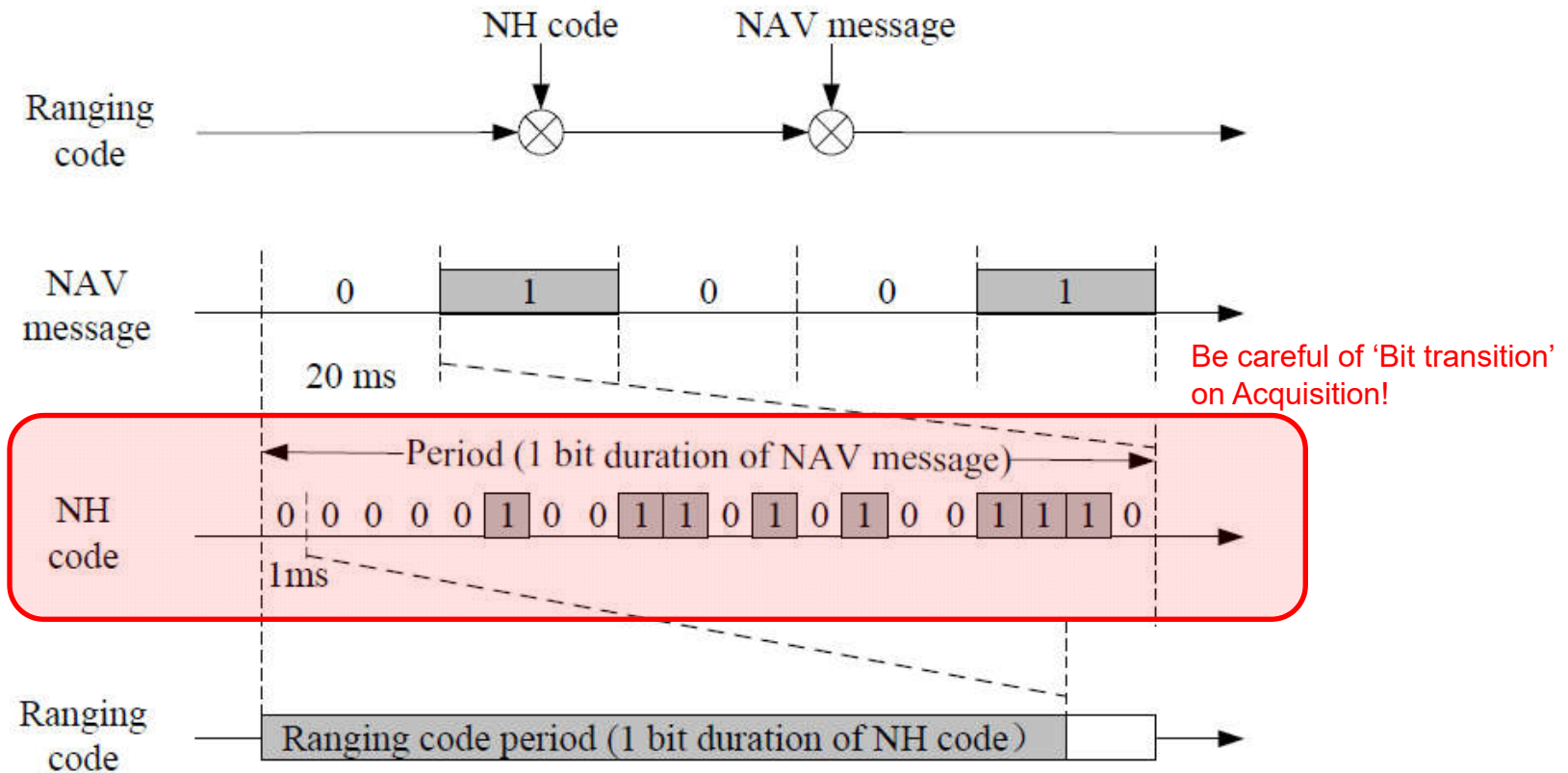
Satellite	Launch (UTC)	Launch site	Carrier Rocket	PRN	Orbit
Compass-G1	1/16/2010	Xichang LC-2	Chang Zheng 3C	C01	GEO 140.0°E
Compass-G3	6/2/2010	Xichang LC-2	Chang Zheng 3C	C03	GEO 110.5°E
Compass-IGSO1	7/31/2010	Xichang LC-3	Chang Zheng 3A	C06	55° inclination IGSO 118°E
Compass-G4	10/31/2010	Xichang LC-2	Chang Zheng 3C	C04	GEO 160.0°E
Compass-IGSO2	12/17/2010	Xichang LC-3	Chang Zheng 3A	C07	55° inclination IGSO 118°E
Compass-IGSO3	4/9/2011	Xichang LC-3	Chang Zheng 3A	C08	55° inclination IGSO 118°E
Compass-IGSO4	7/26/2011	Xichang LC-3	Chang Zheng 3A	C09	55° inclination IGSO 95°E
Compass-IGSO5	12/1/2011	Xichang LC-3	Chang Zheng 3A	C10	55° inclination IGSO 95°E
Compass-G5	2/24/2012	Xichang LC-2	Chang Zheng 3C	C05	GEO 58.75°E
Compass-M3	4/29/2012	Xichang LC-2	Chang Zheng 3B	C11	MEO ~21,500 km, Slot A0 7
Compass-M4	4/29/2012	Xichang LC-2	Chang Zheng 3B	C12	MEO ~21,500 km, Slot A0 8
Compass-M6	9/18/2012	Xichang LC-2	Chang Zheng 3B	C14	MEO ~21,500 km
Compass-G6	10/25/2012	Xichang LC-2	Chang Zheng 3C	C02	GEO 80°E
BDS I1-S	3/30/2015	Xichang LC-2	Chang Zheng 3C/YZ-1	C31	55° inclination IGSO 95°E
BDS M1-S	7/25/2015	Xichang LC-2	Chang Zheng 3B/YZ-1	C33	MEO ~21,500 km, Slot A0 6
BDS M2-S	7/25/2015	Xichang LC-2	Chang Zheng 3B/YZ-1	C34	MEO ~21,500 km, Slot A0 1
BDS I2-S	9/29/2015	Xichang LC-3	Chang Zheng 3B	C32	55° inclination IGSO 95°E
BDS M3-S	2/1/2016	Xichang LC-2	Chang Zheng 3C/YZ-1	C35	MEO ~21,500 km, Slot B0 1
Compass-IGSO6	3/29/2016	Xichang LC-2	Chang Zheng 3A	C13	55° inclination IGSO 95°E
Compass-G7	6/12/2016	Xichang LA-3	Chang Zheng 3C	C17	GEO 144°E

(*) Wikipedia: List of BeiDou satellites

(*) Total 20 satellites are available until May 2017

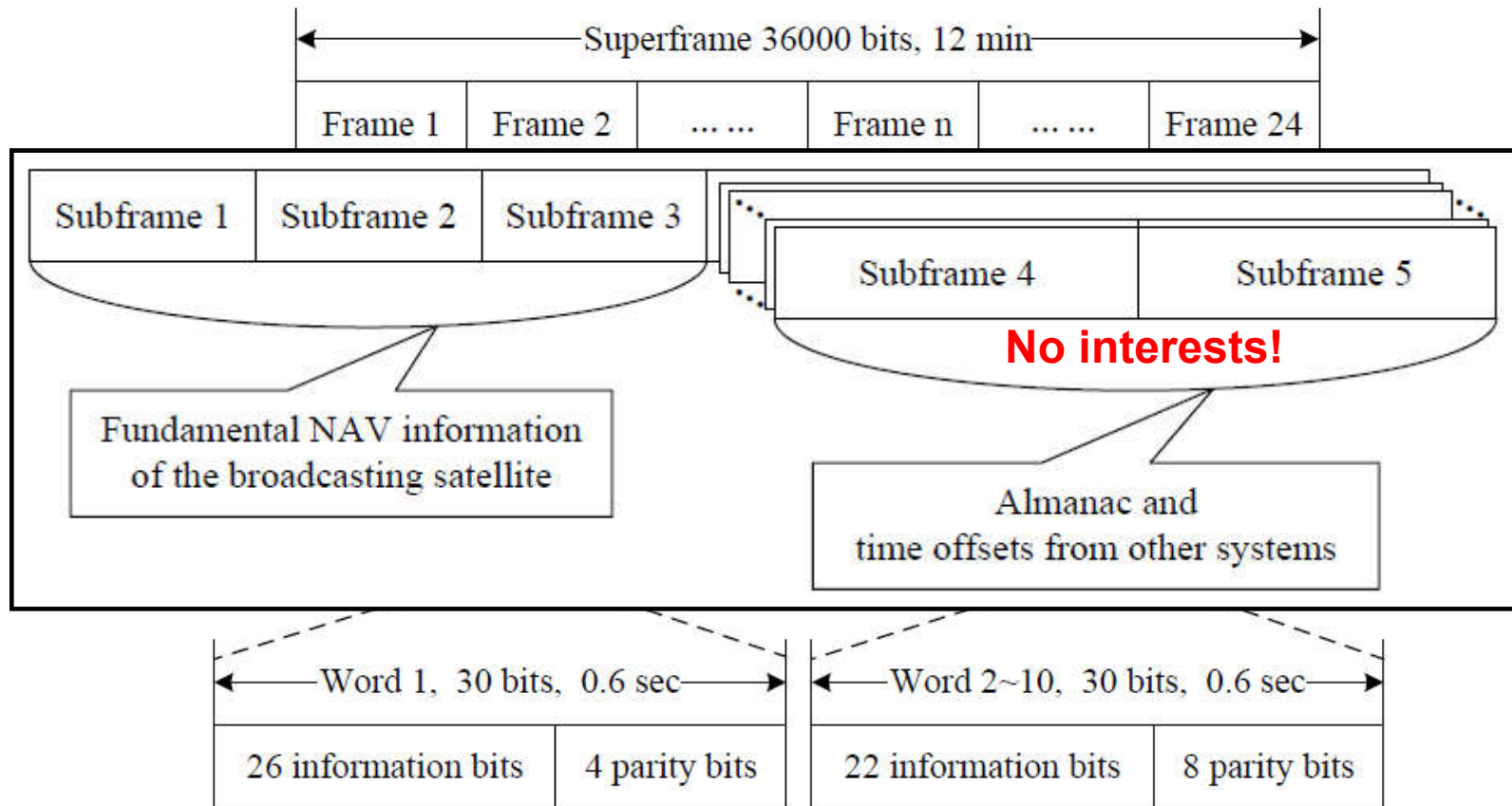
Navigation Message (1/3)

- **D1 Navigation Message (PRN 6–37)**
 - Modulated with 1 kbps *NH code* (20 ms).



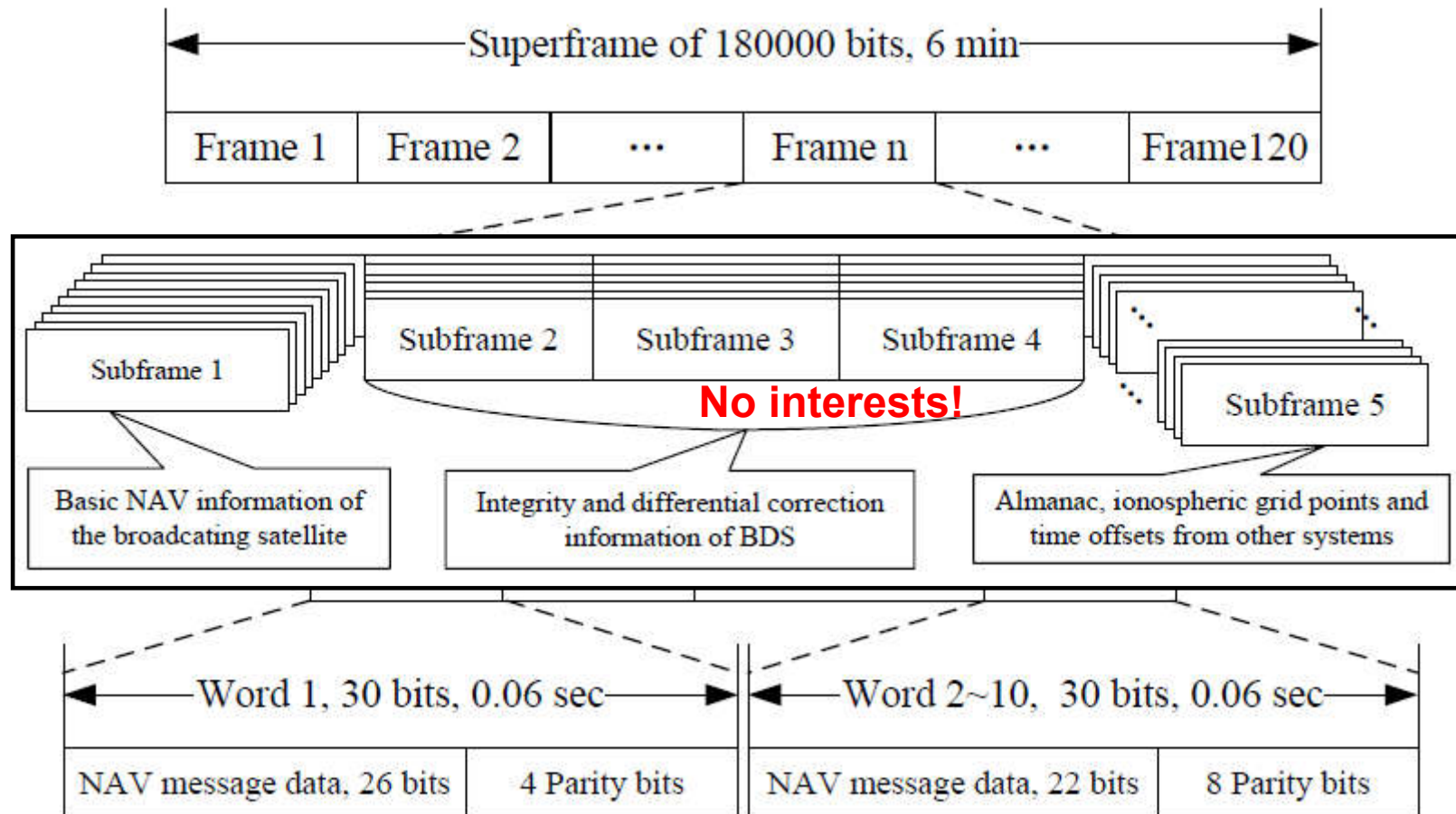
Navigation Message (2/3)

- **Frame Structure of D1 Format (PRN 6–37)**



Navigation Message (3/3)

- **Frame Structure of D2 Format (PRN 1–5)**

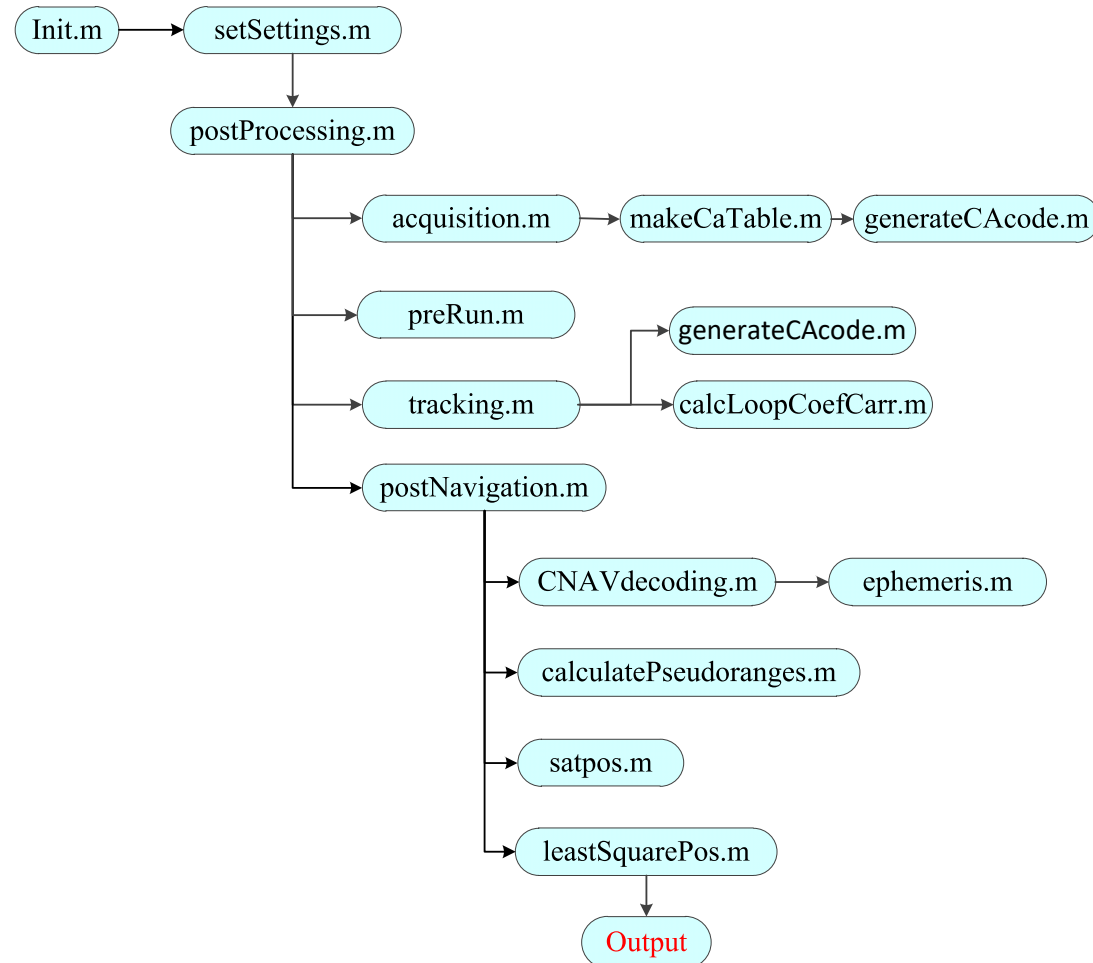


Ephemeris

- **Satellite Position**
 - Use Kepler's equation
 - Ephemeris is almost same as GPS
 - Additional orbital model for GEO

Beidou B1/B2 SDR

SDR Structure



* These are selected blocks to understand easily.

initSetting.m (1/2)

- **File type & sampling information**

- User dependent variables

- This SDR can process both Beidou B1 and B2 open service signals. The only difference for B1/B2 is these variables.

- settings.fileName, settings.dataType, settings.fileType, settings.IF, settings.samplingFreq

- Fixed variables

- Beidou ranging code is 2046 Mcps, and the length is 2046 chips.

- `settings.codeFreqBasis = 2.046e6;`

- `settings.codeLength = 2046;`

initSetting.m (2/2)

▪ Acquisition Setting

- Until May 2017, there are 20 available SVs in space:

`settings.acqSatelliteList = [1:14 17 31:35];`

- Sampling rate threshold for downsampling

- If input signal sampling freq. is too high for FFT computation, it can be resampled to a lower freq. to save processing time.

`settings.resamplingThreshold = 50e6;`

- Enable/disable use of down-sampling for acquisition

`settings.resamplingflag = 1; % 0 – Off, 1 - On`

postProcessing.m

- **Change the length of acquisition data**
 - It depends on the acquisition process. You can modify it.

```
data = fread(fid, dataAdaptCoeff*20*samplesPerCode,  
settings.dataType);
```

acquisition.m (1/2)

- **Selectable down-sampling**
 - Same as that in L2C SDR
- **Create five 4ms vectors of ranging code**
 - Improve **sensitivity** & Cope with “**bit transition**” by NH-code modulation.
 - 4ms vector = [2 x 1ms code, 2ms zeros]
 - ```
caCodeFreqDom = conj(fft([caCodesTable(PRN, :),
 caCodesTable(PRN, :) zeros(1, 2*samplesPerCode)]));
```
  - Cope with the bit transition by NH-code (D1 case).
    - ```
signal1 = longSignal(1 : AcqCodeLength*samplesPerCode);  
signal2 = longSignal(AcqCodeLength*samplesPerCode+1 :  
                      2*AcqCodeLength*samplesPerCode);  
signal3 = longSignal(AcqCodeLength*samplesPerCode*2+1 :  
                      3*AcqCodeLength*samplesPerCode);  
signal4 = longSignal(AcqCodeLength*samplesPerCode*3+1 :  
                      4*AcqCodeLength*samplesPerCode);  
signal5 = longSignal(AcqCodeLength*samplesPerCode*4+1 :  
                      5*AcqCodeLength*samplesPerCode);
```

acquisition.m (2/2)

- Find maximum inverse DFT result

```
[MaxValue, MaxIndex] = max([max(acqRes1) max(acqRes2)  
max(acqRes3) max(acqRes4) max(acqRes5)]);  
switch MaxIndex  
    case 1  
        results(frqBinIndex, :) = acqRes1;  
    case 2  
        results(frqBinIndex, :) = acqRes2;  
    case 3  
        results(frqBinIndex, :) = acqRes3;  
    case 4  
        results(frqBinIndex, :) = acqRes4;  
    case 5  
        results(frqBinIndex, :) = acqRes5;  
end
```

makeCaTable.m

- **Change the number of PRN & the code index.**

```
for PRN = 1:37
```

```
...
```

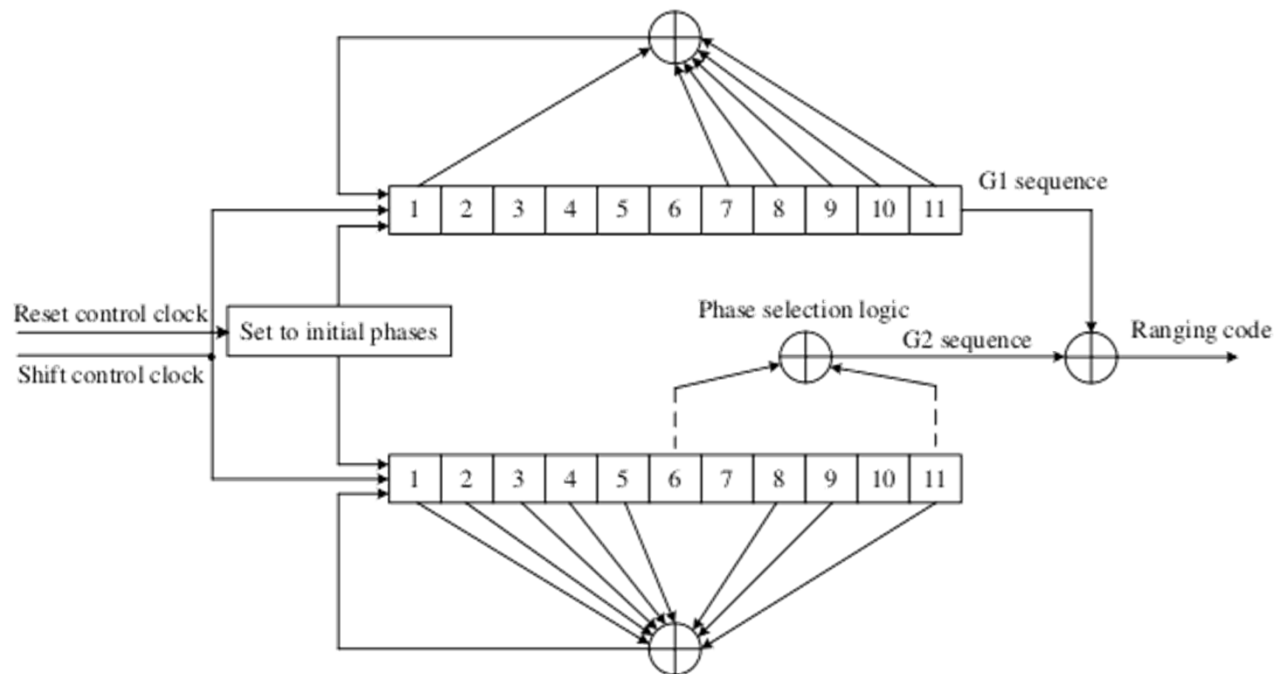
```
    codeValueIndex(end) = 2046;
```

```
...
```

```
end
```

generateCAcode.m (1/2)

- B1I ranging code is generated by means of **Modulo-2 addition of G1 and G2 sequences** which are respectively derived from two 11-bit linear shift registers.
- See “Beidou ICD (Dec. 27, 2012)” for more details.



generateCAcode.m (2/2)

```
function CAcode = generateCAcode(PRN)
g1 = zeros(1, 2046);
reg = -1*[-1, 1, -1, 1, -1, 1, -1, 1, -1, 1, -1];
for i=1:2046
    g1(i) = reg(11);
    SaveBit = reg(1)*reg(7)*reg(8)
              *reg(9)*reg(10)*reg(11);
    reg(2:11) = reg(1:10);
    reg(1) = saveBit;
end
g2 = zeros(1, 2046);
reg = -1*[-1, 1, -1, 1, -1, 1, -1, 1, -1, 1, -1];
g2s1 = [1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 3, ...
         3, 3, 3, 3, 3, 3, 4, 4, 4, 4, ...
         4, 4, 5, 5, 5, 5, 5, 6, 6, 6, ...
         6, 8, 8, 8, 9, 9, 10];
g2s2 = [ 3, 4, 5, 6, 8, 9,10,11, 7, 4, ...
         5, 6, 8, 9,10,11, 5, 6, 8, 9, ...
         10,11, 6, 8, 9,10,11, 8, 9,10, ...
         11, 9,10,11,10,11,11];
for i=1:2046
    g2(i) = reg(g2s1(PRN))*reg(g2s2(PRN));
    saveBit = reg(1)*reg(2)*reg(3)*reg(4)*
              reg(5)*reg(8)*reg(9)*reg(11);
    reg(2:11) = reg(1:10);
    reg(1) = saveBit;
end
CAcode = -(g1 .* g2);
```

tracking.m

- **Not Changed**

postNavigation.m (1/3)

- **Main changes**
 - Differences between D1 & D2
 - NH-code demodulation for D1
 - Different data rate between D1 & D2
 - Nav data decoding (See “NAVdecoding.m”)
 - Ephemeris decoding (See “ephemeris.m”)
 - PVT solution

postNavigation.m (2/3)

■ Main changes

- Start and end of measurement points

- Measurement point begins at the latest start of first messages of all channels (**firstSubFrame**), and consider Doppler smoothing
- **for channelNr = activeChnList**
- **sampleStart(channelNr) = ...**
- **trackResults(channelNr).absoluteSample(firstSubFrame(channelNr));**
- **sampleEnd(channelNr) = trackResults(channelNr).absoluteSample(end);**
- **End**
- **...**

postNavigation.m (3/3)

■ Main changes

- Correct and save local time

- `localTime = localTime - xyzdt(4)/settings.c;`
- `navSolutions.localTime(currMeasNr) = localTime;`

- Update local time

- **Except first time, local time is updated by measurement step**
- `localTime = localTime + measSampleStep/settings.samplingFreq ;`

NAVdecoding.m (1/5)

(1) Bit and frame synchronization

- **Up-sampling the preambles**

- D1: 20 values per one bit with NH-code modulation.
- D2: 2 values per one bit.

```
preamble_bits = [1 1 1 -1 -1 -1 1 -1 -1 1 -1];  
NH_code = [-1 -1 -1 -1 -1 1 -1 -1 1 1 -1 1 -1 1 -1 -1 1  
1 1 -1];  
preamble_D2 = kron(preamble_bits, ones(1, 2));  
preamble_D1 = kron(preamble_bits, -NH_code);
```

NAVdecoding.m (2/5)

(1) Bit and frame synchronization

▪ Set the correlation thresholds

– D1

```
elseif trackResults(channelNr).PRN >= 6 &&  
trackResults(channelNr).PRN <= 37
```

```
    tlmXcorrResult = xcorr(bits, preamble_D1);  
    codrPerD = 20;
```

– D2

```
if trackResults(channelNr).PRN <= 5
```

```
    tlmXcorrResult = xcorr(bits, preamble_D2);  
    codrPerD = 2;
```

NAVdecoding.m (3/5)

(1) Bit and frame synchronization

- **BCH decoding**

% Create a Galois field array from bits

bits = gf(bits,1);

bits = bits(16:30);

% Decode the received signal in code using an

% [15,11] BCH decoder

[~,cnumerr] = bchdec(bits,15,11);

NAVdecoding.m (4/5)

(2) Nav data decoding

▪ D1 decoding

- D1 is modulated by NH-code.
- Each subframe is 300 bits (6 sec), and 1 frame is 1500 bits (30 sec).
- Every 20 raw bits are converted to 1 navigation bit (50 bps).

```
NH_code = [-1 -1 -1 -1 -1 1 -1 -1 1 1 -1 1 -1 1 -1 1 1 1 -1];  
demod_bit = kron(ones(1,300*5), -NH_code);  
...  
elseif trackResults(channelNr).PRN >= 6 && trackResults(channelNr).PRN  
<= 37  
    navBitsSamples =  
    trackResults(channelNr).I_P(subFrameStart(channelNr): ...  
        subFrameStart(channelNr) + (1500 * 20) -1)';  
    navBitsSamples = navBitsSamples' .* demod_bit;  
    navBitsSamples = reshape(navBitsSamples, 20, (size(navBitsSamples,  
2) / 20));  
end
```

NAVdecoding.m (5/5)

(2) Nav data decoding

▪ D2 decoding

- Each subframe is 300 bits (0.6 sec), and 1 frame is 1500 bits (3 sec).
- Every 2 raw bits are converted to 1 navigation bit (500 bps).

```
if trackResults(channelNr).PRN <= 5
    navBitsSamples =
    trackResults(channelNr).I_P(subFrameStart(channelNr):
    subFrameStart(channelNr) + (1500 * 20) - 1)';
    navBitsSamples = reshape(navBitsSamples, 2,
    (size(navBitsSamples, 1) / 2));
    ...
```

ephemeris.m (1/3)

- **Main changes**

- Separated decoding for D1 & D2
 - D1 & D2 have different message structures
 - D1 is similar to GPS NAV message, while D2 is different.
- **Deinterleaving**
 - D1 & D2 are interleaved. So deinterleaving is required for NAV decoding.
- **SOW (Second of week)**
 - First SOWs have different bias.
 - D1: Final SOW - **24** seconds
 - D2: Final SOW - **27** seconds

ephemeris.m (2/3)

- **Deinterleaving (D1 & D2)**

- First word (30 bits) is not interleave.
- MSB 22 bits is information and LSB 8 bits is error check bits.

```
DeSubframe = subframe(1:30);
```

```
for k = 1:9
```

```
    DeSubframe = [DeSubframe, ...  
                  subframe(k*30+1:2:k*30+22), ...  
                  subframe(k*30+2:2:k*30+22), ...  
                  subframe(k*30+23:2:k*30+30), ...  
                  subframe(k*30+24:2:k*30+30)];
```

```
end
```

ephemeris.m (3/3)

- **BCH decoding**

- Decode the received signal in code using an [15,11] BCH decoder.

```
[decoded,cnumerr] = bchdec( gf(subframe(16:30),1),15,11 );
```

calculatePseudoranges.m (1/2)

■ Signal transmitting time of measurement points

- Use method of hardware receiver
- More straightforward and more accurate

$$t_t = TOW + \left(\frac{\text{codePhase}}{\text{code length}} + \text{integer number of C/A code} \right) * 0.001$$

```
· codePhase = trackResults(channelNr).remCodePhase(index) + ...  
· codePhaseStep * (currMeasSample - ...  
· trackResults(channelNr).absoluteSample(index) );  
· transmitTime(channelNr) = (codePhase/settings.codeLength + index - ...  
· subFrameStart(channelNr)) * settings.codeLength/...  
· settings.codeFreqBasis + TOW(channelNr);
```

■ Initialize signal receiving time (local time) t_r

- Use transmitting time and `settings.startOffset`
 - `maxTime = max(transmitTime(channelList));`
 - `localTime = maxTime + settings.startOffset/1000;`

calculatePseudoranges.m (2/2)

- Calculate pseudo-range (PR)

$$PR = (t_r - t_t) * c$$

where, c - light speed

· `pseudoranges = (localTime - transmitTime) * settings.c;`

satpos.m (1/2)

- **Use Kepler's equation**
 - Ephemeris is almost same as GPS
 - Additional orbital model for GEO

satpos.m (2/2)

```
if prn <= 5
```

```
%Compute the angle between the ascending node and the Greenwich meridian  
Omega = eph(prn).omega_0 + eph(prn).omegaDot*tk - Omegae_dot * eph(prn).t_oe;  
%Reduce to between 0 and 360 deg  
Omega = rem(Omega + 2*BeiDouPi, 2*BeiDouPi);
```

```
%--- Compute satellite coordinates -----  
satPositions(1, satNr) = cos(u)*r * cos(Omega) - sin(u)*r * cos(i)*sin(Omega);  
satPositions(2, satNr) = cos(u)*r * sin(Omega) + sin(u)*r * cos(i)*cos(Omega);  
satPositions(3, satNr) = sin(u)*r * sin(i);
```

```
angX = -5*(BeiDouPi/180);  
angZ = Omegae_dot*tk;
```

```
Rx = [1    0    0;  
      0 cos(angX) sin(angX);  
      0 -sin(angX) cos(angX)];
```

```
Rz = [cos(angZ) sin(angZ) 0;  
      -sin(angZ) cos(angZ) 0;  
      0    0    1];
```

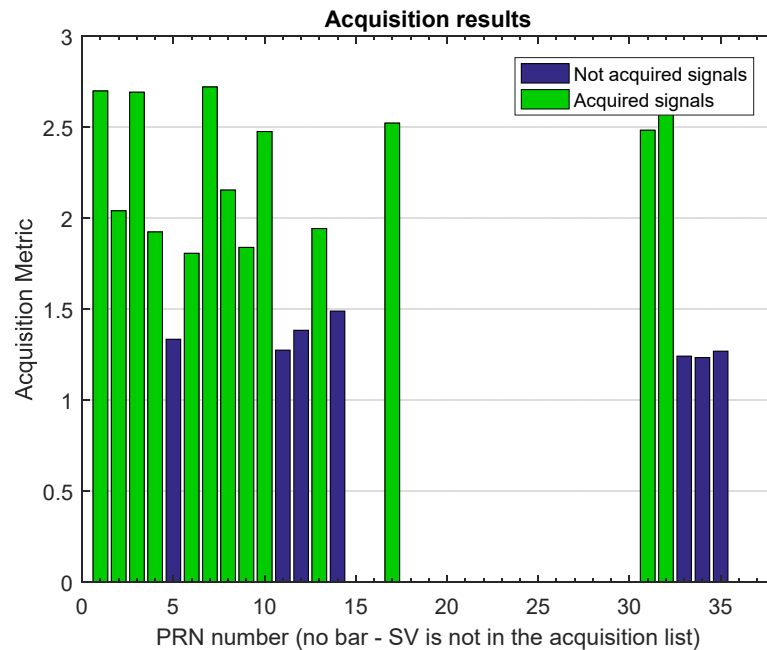
```
satPositions(:, satNr) = Rz * Rx * satPositions(:, satNr);
```

```
...
```

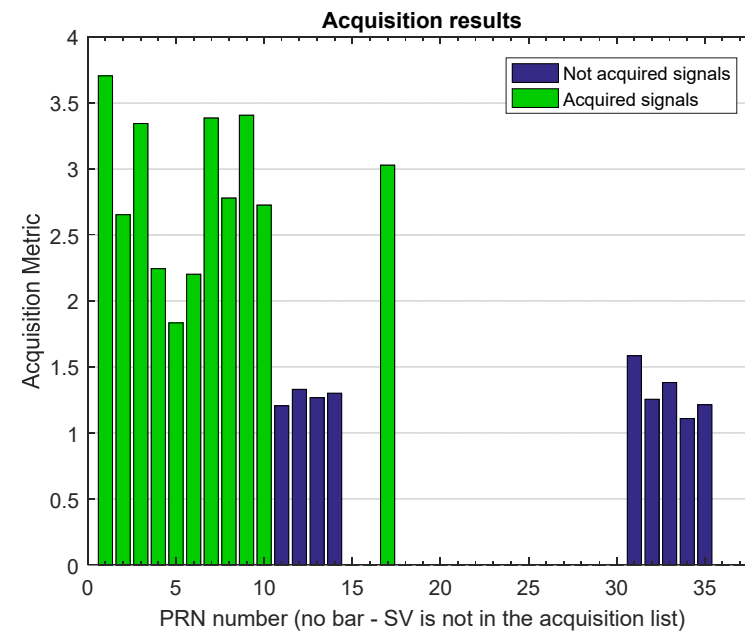
```
end
```

Additional Step for GEO

Acquisition Results (1/1)



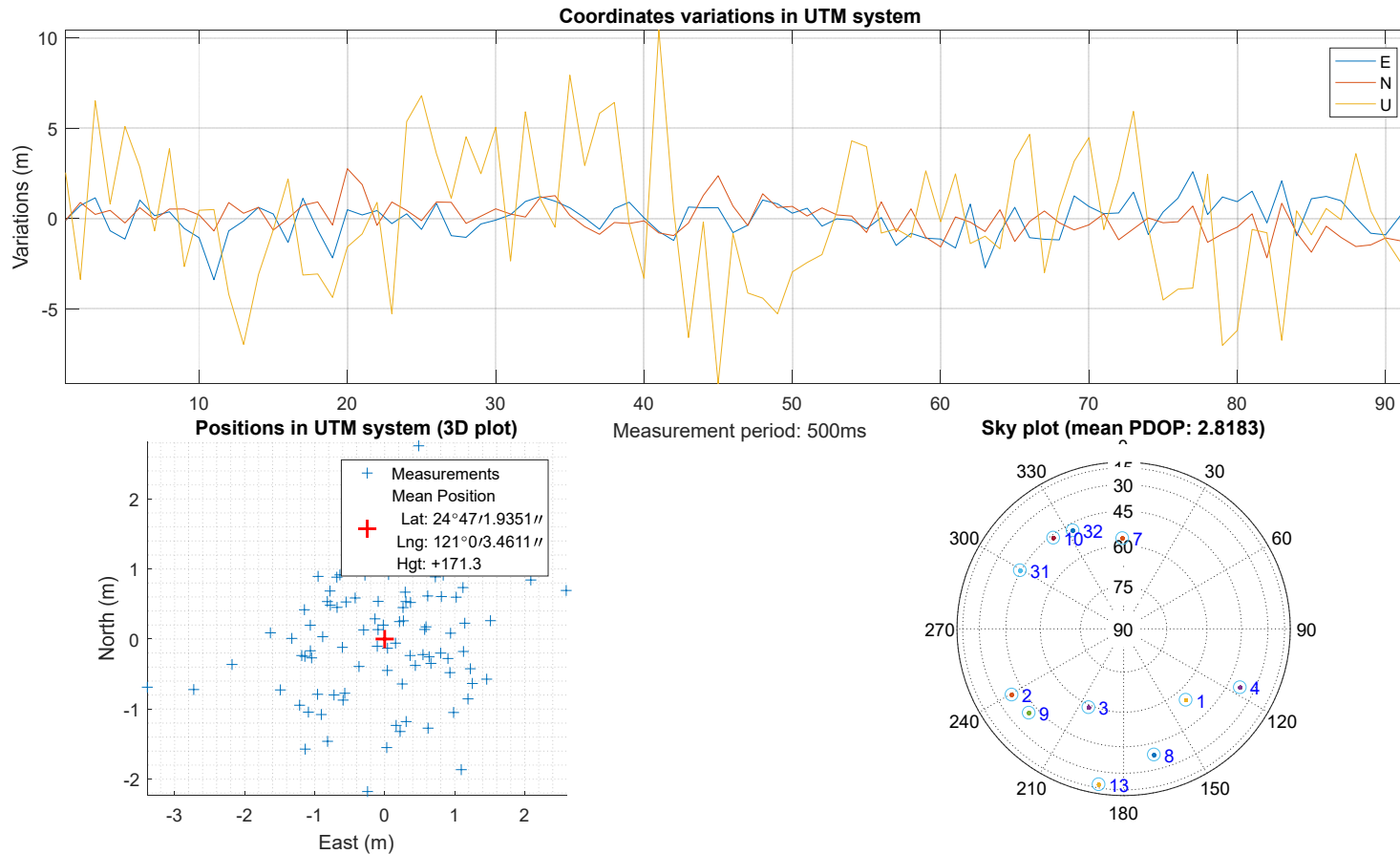
B1I signal



B2I signal

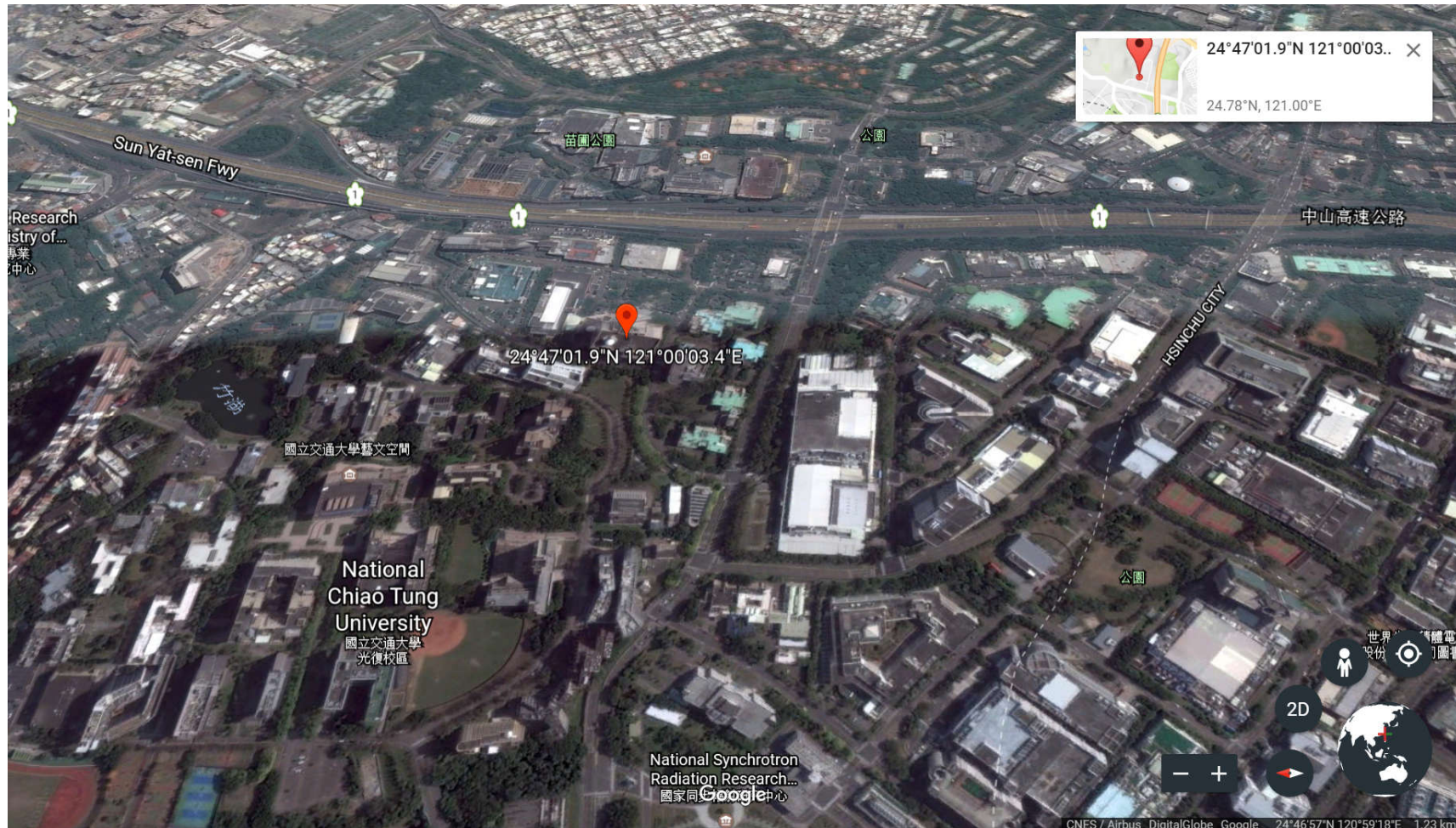
* The IF signal is collected at National Chiao Tung University of Taiwan.

Beidou SDR Results (1/2)



* The IF signal is collected at National Chiao Tung University of Taiwan.

Beidou SDR Results (2/2)



References

- Beidou Navigation Satellite System Signal In Space Interface Control Document (ICD), Open Service Signal B1I (Version 1.0), China Satellite Navigation Office, Dec. 2012.
- Oliver Montenbruck, Peter Steigenberger, “The Beidou Navigation Message”, IGNSS Symposium 2013.
- IGS website: igs.org/mgex/Status_BDS.htm
- GNSS-SDR (*with open source code*):
<http://gnss-sdr.ru/index.php?itemid=89>
- Kai, B., Akos, D. M., Bertelsen, N., Rinder, P., & Jensen, S. H.. A Software-Defined GPS and Galileo Receiver. Birkhäuser Boston, 2007.

Thank you!

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